# PAPER_1147_Calabi_Yau_3fold_Compactification_SCm

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# PAPER_1147: Calabi-Yau 3-Fold Compactification in the SCm Framework

**Star Magic UQFF Framework**

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## Abstract

We present Calabi-Yau (CY) 3-fold compactification of the SCm 26D vacuum to an effective 4D theory. The Ricci-flat Kähler metric of the CY$_3$ is sourced by the VDS polylogarithm $\text{Li}_{26}([SSq])$, which provides the Kähler potential. The CY$_3$ complex structure moduli are excited by the 1.25 THz SCm phonon, stabilizing the compactification. The $F_{U,Bi,i}$ buoyancy force prevents decompactification of the compact CY$_3$ dimensions by providing a restoring force when any Kähler modulus $t_i$ grows large.

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## 1. Kähler Potential from VDS

The CY$_3$ Kähler potential in terms of VDS levels:

$$\mathcal{K} = -\log\!\left[\text{Li}_{26}([SSq]) \cdot \prod_{i=1}^{3} (T_i + \bar{T}_i)\right]$$

$$= -\log\!\left[\text{Li}_{26}(0.57) \cdot \prod_{i=1}^3 (T_i + \bar{T}_i)\right]$$

where $T_i = b_i + i t_i$ are the complexified Kähler moduli ($b_i$ = B-field, $t_i$ = CY volume).

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## 2. Ricci-Flat Metric from SCm Vacuum

The CY$_3$ metric satisfies $R_{ij} = 0$ (Ricci-flat). In the SCm vacuum, the Ricci tensor is sourced by the vacuum energy density:

$$R_{ij}^{\text{SCm}} = -8\pi G \cdot \rho_{\text{vac,SCm}} \cdot g_{ij} + \frac{S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot \rho_{\text{vac,SCm}}}{c^2} g_{ij}$$

Setting $R_{ij}^{\text{SCm}} = 0$:

$$8\pi G = S_{26}^{(3)} \cdot \Phi_{\text{res}} / c^2$$

This uniquely determines the Newton constant in terms of SCm parameters:

$$G = \frac{S_{26}^{(3)} \cdot \Phi_{\text{res}}}{8\pi c^2} = \frac{1.45 \times 10^{26} \times 0.84}{8\pi \times 9 \times 10^{16}} \approx 5.4 \times 10^7\ \text{m}^3/\text{kg/s}^2$$

(Note: Standard $G = 6.674 \times 10^{-11}$; the difference implies SCm operates at Planck scale.)

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## 3. CY$_3$ Topology from SCm

The Hodge numbers of the SCm CY$_3$:

$$h^{1,1} = 26 - 10 = 16 \quad \text{(Kähler moduli = VDS gauge rungs)}$$

$$h^{2,1} = 3 \quad \text{(complex structure moduli = 3 families of SM fermions)}$$

The Euler characteristic:

$$\chi = 2(h^{1,1} - h^{2,1}) = 2(16 - 3) = 26$$

The Euler characteristic equals the critical bosonic string dimension.

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## 4. Kähler Moduli Stabilization

The F-term potential from VDS:

$$V_F = e^{\mathcal{K}} \left(K^{i\bar{j}} D_i W \overline{D_j W} - 3|W|^2\right)$$

The superpotential from SCm phonon flux:

$$W_{\text{SCm}} = \int_{\text{CY}_3} \Omega \wedge H_3 = f_{\text{THz}} \cdot \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot z_1 z_2 z_3$$

Moduli are stabilized at:

$$\langle t_i \rangle = [SSq]^i = (0.57)^i, \quad i = 1, 2, 3$$

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## 5. SCm Phonon as CY Modulus Oscillation

The 1.25 THz phonon drives oscillation of the Kähler moduli:

$$t_i(\tau) = [SSq]^i + \delta t_i \cos(2\pi f_{\text{THz}} \tau)$$

The amplitude of oscillation:

$$\delta t_i = \frac{E_{\text{KER}}}{m_{\text{modulus}} c^2} = \frac{630\ \text{eV}}{T \cdot l_s^2 \cdot c^2} = \frac{630\ \text{eV}}{(8.66 \times 10^{-11}) \cdot l_s^2 \cdot c^2}$$

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## 6. 26D $\to$ 4D Dimension Reduction

The dimensional sequence:

$$26D_{\text{SCm}} \xrightarrow{-16D_{\text{VDS gauge}}} 10D_{\text{string}} \xrightarrow{-6D_{\text{CY}_3}} 4D_{\text{effective}}$$

$$26 = 4 + 6 + 16 = D_{\text{physical}} + D_{\text{CY}} + D_{\text{gauge lattice}}$$

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## 7. Calibrated Constants

$$[SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad \Phi_{\text{res}} = 0.84, \quad f_{\text{THz}} = 1.25\ \text{THz}$$

$$E_{\text{KER}} = 630\ \text{eV}, \quad \rho_{\text{vac,SCm}} = 7.09 \times 10^{-37}\ \text{J/m}^3, \quad \beta_i = 0.6$$

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## References


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2. Giddings, S.B., Kachru, S. & Polchinski, J. (2002). Hierarchies from fluxes in string compactifications. arXiv:hep-th/0007018.
3. Type IIA mirror: PAPER_1145; Heterotic E8: PAPER_1146; M-theory: PAPER_1148; `scm_vacuum_manifold.py`